

Data Analytics

Introduction to Analytics

Turning Data into Decisions

Learn what analytics really is, how it fits with data science and data analysis, and the patterns, tools and techniques that turn data into powerful business insights.



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1 What Really Is Analytics?

Data analytics is the process of collecting, transforming, and organizing data to identify patterns, draw conclusions, and make informed decisions. It involves a range of tools and techniques to turn raw data into actionable insights that can improve business performance, optimize processes, and predict future trends.



The Goal: Turn data into meaningful insights that drive better decisions.

Four Main Types of Analytics

Descriptive



What happened

Diagnostic



Why it happened

Predictive



What is likely to happen

Prescriptive



What should be done

2 Key Processes and Types



Process: Data analytics involves defining objectives, collecting and cleaning data, analyzing it using various techniques, and sharing the results.



Data Cleaning: Critical step to remove duplicates, errors, and missing information from datasets before analysis can begin.



Techniques: Common methods include statistical analysis, regression analysis, time series analysis, data mining, and machine learning.

Types D2P3:

- Descriptive analytics:** Summarizes past data to understand what has happened.
- Diagnostic analytics:** Investigates the root causes of past events.
- Predictive analytics:** Uses historical data to forecast future trends and outcomes.
- Prescriptive analytics:** Recommends a specific course of action to achieve a desired outcome.

3 Common Patterns Identified in Analytics

A dataset can reveal a wide range of patterns, from basic trends to complex relationships. Identifying these patterns is a core goal of data analytics, as it allows analysts to understand past performance, diagnose issues, and predict future outcomes and prescribe solutions.

1. Trend Analysis



Identify general direction over time

2. Seasonal & Cyclical



Find repeating fluctuations

3. Correlation



Understand relationships between variables

4. Clustering



Group similar data points

5. Anomalies



Detect unusual outliers

6. Association

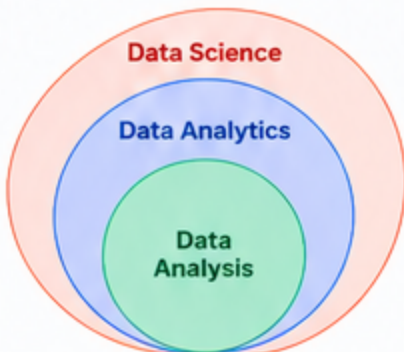


Find items frequently together

4 Analytics vs. Data Science vs. Data Analysis

Related but different — understanding the differences helps you use the right approach.

Aspect	Data Analytics	Data Science	Data Analysis
Main Focus	Insights for better decisions	Building intelligent models and solving complex problems	Examining data to understand what happened
Goal	Actionable business insights	Prediction & automation	Answer specific questions
Techniques	BI tools, visualization, statistical analysis	Machine learning, AI, advanced modeling	Descriptive statistics, lookups, basic analysis
Outcome	Dashboards, reports, decisions	Models, predictions, products	Findings, summaries



Key Takeaway: Analytics is the bridge — using data analysis and data science techniques to turn data into real-world impact.

5 When Analytics Is Involved

- Analytics is involved when the end goal is to turn data into actionable insights — not just to understand the data, but to make better decisions, improve performance, and create future value.

6 Tools & Next Steps



Popular Analytics Tools:

- Microsoft Excel & Access
- Python & R
- Power BI & Tableau
- Minitab (Statistical Analysis)



Your Next Steps:

- Build hands-on skills with real datasets
- Apply analytics to solve business problems
- Learn how analytics connects with data science
- Grow into data-driven decision making



Data drives the future. Master analytics to turn information into opportunity.

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